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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

EE2102-1- ELECTRICAL MACHINES - I
(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Transformer core is constructed for-----**
A) Providing least effective magnetic linkage between two windings
B) Providing isolation between magnetic linkages of one coil from another
C) Providing most effective magnetic linkage between two windings
D) Cannot be determined
2. **Transformer cores are laminated in order to**
A) simplify its construction
B) minimize eddy current loss
C) reduce cost
D) reduce hysteresis loss
3. **The primary and secondary induced e.m.fs. E1 and E2 in a two-winding transformer are always**
A) equal in magnitude
B) in-phase with each other
C) antiphase with each other
D) determined by load on transformer secondary
4. **In a two-winding transformer, the e.m.f. per turn in secondary winding is always.....the induced e.m.f. per turn in primary.**
A) equal to
B) equal to K times
C) equal to 1/K times
D) greater than
5. **In the equivalent circuit of the transformer, we include reactance to represent the effect of**
A) Resistance of the winding
B) Laminations of transformers.
C) Rate of change of flux
D) Leakage flux
6. **A transformer having 1000 primary turns is connected to a 250-V a.c. supply. For a secondary voltage of 400 V, the number of secondary turns should be**
A) 1600
B) 250
C) 400
D) 1250
7. **The equivalent resistance of the primary of a transformer voltage ratio of 1000/500V and $R_1 = 0.1$ ohm when referred to secondary becomes _____ ohm.**
A) 0.4
B) 0.04
C) 0.025
D) 0.25
8. **The winding arrangement in a shell type transformer is referred as-----**
A) Concentric winding
B) Cylindrical winding
C) Helical winding
D) Sandwich winding

9. Of the following statements concerning parallel operation of transformers, the one which is not correct is
- A) transformers must have equal voltage ratings
B) transformers must have same ratio of transformation
C) transformers must be operated at the same frequency
D) transformers must have equal kVA ratings
10. In an autotransformer, energy transformation takes place by
- A) Radiation
B) Induction and conduction
C) Induction
D) conduction
11. ----- is an advantage of skewing of cage rotor conductors.
- A) Uniform Torque
B) Non uniform Torque
C) Reduces the air gap between stator and rotor
D) Reduced eddy current loss
12. Difference between the speeds of rotating magnetic field and rotor is called as
- A) Airgap speed
B) Slip speed
C) Synchronous speed
D) Relative speed
13. Condition for maximum torque in 3-phase induction motor is
- A) $X_2 = sR_{20}$
B) $R_2 = 2X_{20}$
C) $R_2 = X_{20}$
D) $R_2 = sX_{20}$
14. The no load current in 3-phase induction motor varies depending on its size in the range of ----
- A) 2 to 5%
B) 25 to 40%
C) 10 to 20%
D) 50 to 60%
15. In per phase equivalent circuit of 3-phase induction motor referred to stator, mechanical load is represented as,
- A) $R'_2 \left(\frac{1-s}{s} \right)$
B) $R'_2 \left(\frac{s}{1-s} \right)$
C) $\frac{R'_2 s}{s}$
D) $\frac{R'_2}{s}$
16. Starters are used in 3-phase induction motors for the purpose of -----
- A) To produce torque at zero speed
B) To produce torque at nonzero speed
C) To reduce heavy starting current
D) To overcome initial friction
17. Rotor resistance speed control method is used in ----- type of motor
- A) Squirrel cage induction motor
B) Capacitor start 1-phase induction motor
C) Slip ring Induction Motor
D) Double cage induction motor
18. Double cage induction motors have the feature of -----
- A) High starting torque at low starting current
B) High starting torque at high starting current
C) Low starting torque at low starting current
D) Low starting torque at high starting current
19. Conventional squirrel cage motors suffer from the disadvantage of ----- starting torque because of ----- rotor resistance.
- A) low, high
B) low, low
C) high, high
D) high, low
20. Single phase induction motors develop ----- torque at slip(s)=1.
- A) minimum
B) maximum
C) rated
D) zero

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the construction of shell and core type of single-phase transformers	06	L*2	1	1,2,12, PSO2

- b) Prove that for a given kVA rating, copper is saved in autotransformer compared to two winding transformers. 6 L4 2 1,2,12, PSO2
- c) A 100 kVA, 6600/440 V, 50 Hz single phase transformer has 80 turns on the low voltage winding. Calculate i) maximum flux in the core, ii) number of turns on the high voltage winding and iii) rated current in each winding 4 L3 1 1,2,12, PSO2
2. a) With necessary circuit diagram and phasor diagram explain delta-delta three phase transformer connection. List two advantages of this. 5 L3 2 1,2,12, PSO2
- b) With necessary circuit diagram explain OC-SC test and explain the method of determining equivalent circuit parameters of transformer from OC-SC test results. 6 L3 1 1,2,12, PSO2
- c) A three-phase transformer is used to step down the voltage of a 3-phase 11 kV feeder line. Per phase turns ratio is 12. For a primary line current of 20 A, calculate the secondary line voltage, line current and output kVA for following connections.
- i) Star-delta
- ii) Delta-delta 5 L4 2 1,2,12, PSO2
3. a) Using necessary equations, explain the separation of hysteresis and eddy-current losses from the iron losses of the transformer. 4 L2 1 1,2,12, PSO2
- b) A 100kVA, 2000/200V, 50 Hz distribution transformer has core loss of 500W at rated voltage and copper loss of 1200W at full load, it has the following load cycle:
- | % load | 0% | 50% | 75% | 100% | 110% |
|--------|----|-----|---------|----------|------|
| p.f. | - | 1 | 0.8 lag | 0.85 lag | 1 |
| Hours | 3 | 6 | 8 | 5 | 2 |
- Determine all day efficiency of the transformer. 6 L3 1 1,2,12, PSO2
- c) List the six applications of three winding transformer. 6 L1 2 1,2,12, PSO2

Unit – II

4. a) Develop the expression for developed torque in a 3-phase induction motor and derive the condition for maximum developed torque. 8 L3 4 1,2,12, PSO2
- b) A 3-phase slip ring induction motor gives a reading of 60 V across the slip rings when at rest with normal stator voltage applied. The rotor is star connected and has impedance of $(0.8+j6) \Omega$ per phase. Find the rotor current when the machine is,
- i) At standstill with the slip rings joined to a star-connected starter with a phase impedance of $(4+j3) \Omega$
- ii) Running normally with 6% slip 8 L2 3 1,2,12
5. a) Prove that when a balanced three phase supply is provided to the stator of a 3-phase induction motor, a rotating magnetic field is produced. 8 L3 3 1,2,12
- b) A 10 kW, 400 V, 4-pole delta connected squirrel cage induction motor gave the following test results.
No load test: 400 V, 8A, 250 W
90 V, 35A, 1350 W
The dc resistance of the stator winding per phase measured immediately after the blocked rotor test is 0.6Ω .
Calculate the rotational losses and equivalent parameters 8 L3 4 1,2,12, PSO2

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|-------|---|---|----|---|-----------------|
| 6. a) | With necessary circuit diagram explain self-excited or isolated induction generator. | 8 | L3 | 4 | 1,2,12,
PSO2 |
| b) | With necessary phasor diagram explain the power factor of 3-phase induction motor at no load and full load. | 8 | L2 | 3 | 1,2,12 |

Unit – III

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|-------|--|---|----|---|-----------------|
| 7. a) | With necessary circuit diagram, phasor diagram and torque speed characteristics explain split phase induction motor and capacitor start motor. | 8 | L3 | 5 | 1,2,12,
PSO2 |
| b) | Explain double- revolving theory of single-phase induction motors. | 8 | L3 | 5 | 1,2,12,
PSO2 |
| 8. a) | The standstill impedance of the outer cage of a double cage induction motor is $(0.3+j 0.4) \Omega$ and that of inner cage is $(0.1+j1.5) \Omega$. Compare the relative currents and torques of two cages (i) at standstill (ii) at slip of 5%. Neglect stator winding impedance. | 8 | L2 | 5 | 1,2,12,
PSO2 |
| b) | Explain the effect of space harmonics on the performance of three-phase induction motor. | 8 | L2 | 5 | 1,2,12,
PSO2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
